

Room 13 (P1 N=6 escape routes): status note, not attempted properly

2026-09-27

room11_seat_erdos.tex names two escape routes from the saddle-pinned-resonance obstruction: (a) a different contour decomposition whose constant-height legs avoid the resonance angle – already assessed there as “not obviously possible,” since the resonance angles are rigidly tied to $\zeta = e^{i\pi/3}$ itself; (b) a genuinely new, $\theta = 0$ -adapted replacement for the ray lemma, giving a lower bound on $|1 - w|$ using only modulus decay, valid *along a path lying exactly on the resonance line*, strictly harder than the already-unattempted generic modulus-only rescue in P1_N6_note.tex.

Attempted here: a quick sanity check of (b), found unreliable and not reported as a result. A five-minute numerical probe (moving x along the real direction from the $N = 6$ saddle x_{-1} and tracking $w = ce^{-x}$) found $\arg w$ constant at 90° throughout, suggesting w never approaches the real axis and $|1 - w|$ stays comfortably large – but this used $w = ce^{-x}$ directly rather than the lemma’s actual resonance-relative quantity (built from one of the five specific $c_j = \zeta_6^i$ values, not c itself), so it is very likely testing the wrong object. **This is flagged explicitly as an unreliable probe, not a finding**, to avoid introducing a false lead of the kind this session has already caught and corrected several times. A proper attempt at route (b) needs the exact resonance-relative w from lem:qi-bounds’s actual statement, re-derived carefully for the $N = 6$ saddle-pinned path, which was not done here.

Verdict: NOT ATTEMPTED (route a is closed off in room11’s own analysis; route b needs a careful, properly-scoped attempt that this quick check did not provide). Both routes remain genuinely open for a future session with a fresh, careful pass at the exact formulas.